

PROFESSIONAL SUMMARY

Automotive Systems Engineer experienced in BCM system requirements elicitation, and specification in compliance with ASPICE SYS.1/SYS.2. Experienced in BCM feature development, cross-functional coordination, and system-level lifecycle activities from requirements to ECU release.

WORK EXPERIENCE

Aptiv Components India Pvt Ltd

Associate Systems Engineer

Bengaluru, India

Jul 2024 – Present

- Worked as a Systems engineer for **Body Control Module (BCM)**, coordinating with software, validation, manufacturing, program, and supplier teams to align system deliverables across project milestones.
- Performed system **requirement elicitation** (ASPICE SYS.1) and system/product requirements specification(ASPICE SYS.2) for BCM features, including **Vehicle Access & Security (PEPS)**, **Power Windows and Mirrors**, **Exterior and interior lighting** and **horn** functions and **Power Management (Sleep/Wake-up)**, using **IBM DOORS** and **Polarion** .
- Owned the **system requirements lifecycle**, including **baseline**, **bidirectional traceability** across V-model, managed **change requests** received from OEM, KPI tracking, and **backlog alignment**.
- Managed **ECU release** preparation activities, including **MPI** creation and validation (CRC verification, variant consistency), coordination with manufacturing teams for flash readiness validation, and execution of **controlled release workflows (SWIT)**.
- Performed system checkout and issue analysis for the BCM, including log and trace analysis, UDS diagnostic verification, and coordination with hardware and software teams during defect investigation.
- Contributed to system-level functional safety activities and DFMEA for the BCM in compliance with **ISO 26262**, also derived Functional Safety Requirements for BCM features.
- Performed system-level requirement analysis and specification related to **Software Authentication** and **Secure Boot** for the BCM, ensuring alignment with defined platform Cybersecurity requirements.

PROJECTS

- **Eye-Blink Controlled Wheelchair** - Developed a hands-free mobility system using IR sensors for eye-blink detection. Implemented blink-pattern recognition, safety interlocks; shortlisted for **YUKTI-NIC 2023** and **Smart India Hackathon (SIH) 2023**.

CORE SKILLS

- **Systems Engineering:** Requirements elicitation, and specification (ASPICE SYS.1 / SYS.2), bidirectional traceability using V-model, change request and system checkout.
- **Automotive Protocols:** CAN, LIN, UDS(ISO 14229): basic FlexRay exposure
- **Tools:** IBM DOORS, Polarion, JIRA, CANoe, HexView
- **ECU Release & Manufacturing :** MPI creation and validation, release readiness coordination, controlled SW release workflows (SWIT)
- **Programming:** C, C++, Python

EDUCATION & INTERNSHIP

Aptiv Components India Pvt Ltd

System Engineering Intern

Bengaluru, KA

Jan 2024 – Jun 2024

G Narayannama Institute of Technology and Science

BTech in Electronics and Telematics, CGPA: 8.37/10

Hyderabad, TS

2024